

GLOBAL INDUSTRY CHALLENGE 2026

TEAM COVER PAGE

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Challenge Name: Quantum-Enhanced Strategic Siting of Energy Storage and Microgrids				
Phase #: 3				
Team Name: Entangled Trio				
First Name	Last Name	Email	Aqora Username	Role within Team
Niraj	Venkat	venkatn93@gmail.com	nirajvenkat	Project Manager / Team Coordinator
Mackenson	Polché	polchemackenson@yahoo.com.mx	mack_polche	Domain Expert
Muskan	Aman	www.muskanaman@gmail.com	muscaanmnmnm	Data / Technical Lead

1 Focus Area, Rationale, and Stakeholder Relevance

This project addresses the strategic siting and capacity sizing of clean energy microgrids and storage systems within power distribution grids. Siting decisions represent a high-stakes, combinatorial planning challenge under complex electrical constraints. To guide real planning outcomes, we model a multi-objective optimization problem that co-optimizes grid resilience (C_R), capital installation cost (C_I), and operational voltage control (C_{VC}). Our pre-rendered Jupyter notebooks on siting optimization, AI load scaling and $N - 1$ resilience are available at [1, 2].

The primary stakeholders and beneficiaries of this methodology include:

- **Electric Utilities and ISOs:** Enables optimal capital allocation for grid assets and helps maintain power quality under dynamic load conditions.
- **Department of Energy (DOE):** Accelerates grid decarbonization and modernization mandates through quantum-enhanced optimization frameworks.
- **Local Communities:** Enhances energy security and resilience against catastrophic outages during extreme weather events.

2 Technical Methodology and Hybrid Quantum Integration

In our **first** notebook, we run a hybrid Benders-QAMOO workflow on IEEE-33 bus. Our approach combines Quantum Approximate Multi-Objective Optimization (QAMOO) by [3] with classical Benders decomposition for strategic siting of energy storage and microgrids. Our hybrid workflow is inspired by [4], but with a different architecture partition: it divides the optimization into a classical Master Problem (MP) and an AC power-flow Subproblem (SP). At its core, this is formulated as a mixed-integer linear programming (MILP) optimization problem, which is loaded into a Qiskit QuadraticProgram. We jointly optimize the following objectives inspired by [5]: C_I : capital investment cost of siting the microgrid assets, C_R : resilience of the grid, and C_{VC} : voltage control quality. This takes the following minimization: $\min_{\bar{x} \in \{0,1\}^N} C(\bar{x}) = \lambda_I C_I + \lambda_R C_R - \lambda_{VC} C_{VC}$

In our **second** notebook, we execute a QAOA pipeline against non-linear structural shocks, we stress-tested IEEE 39 bus across two severe grid-modernization scenarios: an additive load injection model simulating hyperscale AI data centers, and an exhaustive contingency sweep across all 46 transmission lines.

2.1 Classical Benders Preprocessing (CPU)

The MP handles the discrete layout coordinates, selecting the binary portfolio of bus locations for asset placement ($\hat{x}^{(t)} \in \{0, 1\}^N$). It is solved using CPLEX [6]. The candidate layout $\hat{x}^{(t)}$ is passed to the SP evaluator, which runs an AC power-flow simulation using pandapower [7]. If physical line limits (thermal branch loading $> 100\%$) or voltage limits ($0.95 \leq V_j \leq 1.05$ p.u.) are violated, the SP generates linear feasibility cuts:

- **Line Overload Cuts:** For overloaded branches between bus u and v : $x_u + x_v \leq 1$.
- **Voltage Deviation Cuts:** For bus w experiencing operational bounds violation: $x_w \leq 0$.
- **Exact Vector Forbid (Fallback):** If specific buses cannot be isolated: $\sum_{i:\hat{x}_i^{(t)}=1} x_i + \sum_{i:\hat{x}_i^{(t)}=0} (1 - x_i) \leq N - 1$.

These cuts compile iteratively ($T = 5$) to form a set of linear constraints on the master layout solver.

2.2 QUBO Penalty Mapping

To embed feasibility constraints into the QAMOO quantum layer, we convert the collected cuts into quadratic penalty terms $p_i = \lambda \cdot \hat{x}_i^{(t)}$ that modify the self-loop diagonals of the target Ising objective graphs, where $\lambda = 10.0$ is the penalty weight. This mutates the diagonal terms of the objective graphs to penalize infeasible portfolios, steering the quantum variational sampler away from voltage and thermal limit violations.

2.3 Quantum Optimization and Error Mitigation

Once the penalized graphs are compiled, we train a multi-objective QAOA ($p = 3$ layers) optimized using COBYLA, giving us a warm-start on a scalarized representation of the three objectives. The Pareto front is evaluated by executing batched sample runs of size $S = 10$ using Dirichlet-distributed scalarization weights.

To overcome physical device noise, we have implemented and integrated the client-side Samplomatic error-mitigation pipeline using Qiskit’s new Executor primitive. This pipeline combines three error mitigation techniques: Probabilistic Error Cancellation (PEC), Shaded Lightcones (SLC) and Propagated Noise Absorption (PNA).

2.4 AI Data Center Shock Modeling and Multi-Objective $N - 1$ Contingency Resilience

We implement an additive load framework simulating hyperscale AI campus deployment by injecting independent point-source demands (+300 MW / +90 MVar) directly onto two buses. We track voltage boundary violations across the network. When nominal asset allocations fail, the pipeline dynamically reallocates storage assets to the newly vulnerable corridor to support the local voltage floor.

Additionally, we model cost penalties with an empirical resilience metric (replacing C_I for $C_{\text{resilience}}$). We model this by simulating a complete outage tree by sequentially taking all 46 lines out of service. Non-linear grid stability responses under each failure are compiled into a per-bus resilience sensitivity index (R_n). This index mutates the self-loop diagonals of the optimization QUBO as follows: $Q_{i,i} = -\alpha V_n[i] + \mu r_n[i] - \nu R_n[i] + \lambda(1 - 2K)$

3 Data Sources and Modeling Assumptions

This section documents the provenance of all external datasets used in the Phase 3 pipeline, explicitly delineating what is derived from real-world government data versus what constitutes a modeling assumption.

3.1 Data Source Provenance

3.2 AI Data Center Load Injection Model

What is from real data. The total megawatt addition for the TVA Balancing Authority under the high-growth scenario ($\Delta P_{\text{DC}} = 383.40$ MW) is sourced directly from the PNNL/EPRI TELL dataset [8]. This dataset provides hourly Balancing Authority–level load projections with and without data center demand under moderate and high growth scenarios, covering all 66 U.S. BAs.

What is a modeling assumption. The TELL dataset operates at the BA level and does *not* specify which substations within a BA absorb data center demand. Since IEEE benchmark grids are synthetic topologies with no geographic

Table 1: External Dataset Provenance

Pipeline Input	Source Dataset	DOI / ID	Provenance
Total AI load addition (ΔP_{DC})	PNNL/EPRI IM3+EPRI Data Center Load Projections (TELL BA)	10.57931/3007669	Real govt. data
Outage event category probabilities $P(\text{cat})$	PNNL Event-Related Outage Dataset (EAGLE-I $\times$ DOE-417, 2014–2023)	OEDI 6458	Real govt. data
Bus-level AI load allocation (which buses receive ΔP_{DC})	Degree-centrality hub selection with 15% concentration ratio	—	Modeling assumption
Line hazard categorization (Weather / Physical / Equipment)	Thermal loading and degree-centrality percentile thresholds	—	Modeling assumption
IEEE bus test systems (14/33/39/118)	pandapower standard library [7]	—	Synthetic benchmark

mapping, any sub-BA bus-level allocation is necessarily a modeling choice.

We adopt a **sparse hub-bus injection model** based on the following empirical evidence of geographic clustering:

- **PJM (2025):** Data centers account for 94% of projected load growth, with demand concentrated in the Northern Virginia corridor (“Data Center Alley”) [9].
- **IEA (2025):** Approximately 50% of U.S. data centers under development cluster near existing large facilities at high-voltage interconnection points [10].
- **LBNL Queued Up (2025):** Approximately 80% of large-load interconnection requests target fewer than 15% of available substations [11].
- **WECC (2024):** Data center interconnection requests are geographically concentrated at a small number of high-capacity substations [12].

Based on these sources, we set $f_{\text{hub}} = 0.15$ (15% of load buses) as the concentration fraction. This value is consistent with the LBNL finding that <15% of substations absorb the majority of large-load requests. The selected hub buses are the top- $\lceil f_{\text{hub}} \cdot N_{\text{load}} \rceil$ buses ranked by graph degree centrality (a proxy for high-voltage interconnection capacity). Within the selected hub set, demand is distributed proportionally to existing bus load:

$$P_n^{\text{shocked}} = P_n^{\text{base}} + \frac{P_n^{\text{base}}}{\sum_{k \in \mathcal{H}} P_k^{\text{base}}} \cdot \Delta P_{DC}, \quad n \in \mathcal{H} \quad (1)$$

where $\mathcal{H}$ is the set of hub buses and $P_n^{\text{base}} = 0$ for $n \notin \mathcal{H}$ (no injection at non-hub buses).

To ensure AC power flow convergence on small test grids (IEEE-14, IEEE-33), we additionally cap the total injection at 25% of baseline system capacity: $\Delta P_{DC}^{\text{eff}} = \min(\Delta P_{DC}, 0.25 \cdot \sum_k P_k^{\text{base}})$.

3.3 $N - 1$ Contingency Resilience Model

What is from real data. The empirical event-category probabilities used to weight the $N - 1$ contingency metric ($dist_n$) are derived from the PNNL Event-Related Outage Dataset [13], which synthesizes 10 years (2014–2023) of EAGLE-I county-level outage records with DOE-417 Electric Emergency Incident and Disturbance Reports, covering 663 major grid disturbance events across the continental United States. The resulting empirical probabilities are:

Table 2: Empirical Outage Category Probabilities (OEDI 6458, 2014–2023)

Event Category	$P(\text{cat})$	Source
Weather / Natural Disaster	44.9%	EAGLE-I $\times$ DOE-417
Physical / Security	27.3%	EAGLE-I $\times$ DOE-417
Equipment / Operational	24.6%	EAGLE-I $\times$ DOE-417
Other / Unknown	3.0%	EAGLE-I $\times$ DOE-417
Cyber	0.2%	EAGLE-I $\times$ DOE-417

What is a modeling assumption. The mapping from hazard categories to specific transmission lines within IEEE benchmark grids is a modeling assumption, since these synthetic topologies have no geographic or physical-asset metadata. We use a **physics-grounded percentile categorization** scheme that generalizes across any grid size:

- **Weather hazards** $\rightarrow$ Lines in the top-20% thermal loading percentile (proxy: exposed, high-stress corridors).
- **Physical/Security hazards** $\rightarrow$ Lines touching hub buses in the top-15% degree centrality percentile (proxy: high-value interconnection targets).
- **Equipment/Operational hazards** $\rightarrow$ Remaining lines (proxy: routine failure modes).

For each category, a single representative line (the highest thermally loaded within its category) is taken out of service ($N - 1$ contingency), and reactive power probes (± 5.0 MVAR) at each candidate bus measure voltage recovery improvement (R_n^{cat}). The final per-bus resilience metric is:

$$dist_n = \sum_{\text{cat}} P(\text{cat}) \cdot \frac{R_n^{\text{cat}}}{\max_k R_k^{\text{cat}}} \quad (2)$$

3.4 AC Power Flow Solver Robustness

Under stressed loading conditions (AI load shock + $N - 1$ outage), the standard Newton-Raphson (NR) AC power flow solver may fail to converge. We implement a cascading fallback strategy: NR (max_iter = 100) $\rightarrow$ Iwamoto Damped NR $\rightarrow$ Backward/Forward Sweep (BFSW). This ensures convergence across all test cases without masking genuine physical infeasibility.

3.5 Summary of Modeling Assumptions

4 Data Modeling and Benchmarking Strategy

We primarily focus on IEEE datasets. Benders-QAMOO is benchmarked against two exact classical methods:

1. **CPLEX Weighted-Sum:** Classical solver executing scalarized MILP evaluations.

Table 3: Summary of Key Modeling Assumptions and Justifications

Assumption	Value / Choice	Justification
Hub concentration fraction f_{hub}	15% of load buses	LBNL Queued Up 2025: 80% of requests at <15% of substations
Hub selection criterion	Degree centrality	Proxy for high-voltage interconnection capacity
Max load shock cap	25% of baseline ΣP	Power flow convergence on small IEEE grids
Weather line threshold	80th pctl. thermal loading	Exposed, high-stress corridors
Physical/Security bus threshold	85th pctl. degree centrality	High-value interconnection targets
$N - 1$ representative line	Max loading per category	Tractable probing (1 line/category vs. exhaustive)

2. **Defining Point Algorithm (DPA):** State-of-the-art MILP algorithm with provable convergence.

5 Experimental Results and Discussion

First Notebook: Table 5 measures key multi-objective optimization metrics: Pareto front size ($|\text{PF}|$), Hypervolume (HV), Inverted Generational Distance (IGD), spacing, additive epsilon indicator (ϵ^+), and coverage ($C(A, B)$).

Table 4: Pareto Quality Metrics on IEEE 33-Bus Grid

Method	PF	HV	IGD	Spacing	ϵ^+	Coverage	
						$C(A, B)$	$C(B, A)$
QAMOO (Quantum)	189	707.2790	0.0961	0.3724	0.0588	0.5455	0.0529
Benders-QAMOO (Quantum)	270	824.3916	0.1097	0.2613	0.0714	0.4697	0.0333
CPLEX (Exact Classical WS)	9	47.6514	0.1466	0.1560	0.2569	0.1818	1.0000
DPA (Exact Classical)	57	49.1532	0.0001	0.0665	0.0004	0.8636	1.0000

The empirical metrics reveal a stark architectural trade-off between quantum exploration and classical precision. The quantum models demonstrate massive dominance in exploring non-dominated design spaces, with Benders-QAMOO producing a Pareto front of size $|\text{PF}| = 270$. This dense coverage yields an order-of-magnitude superiority in Hypervolume (HV), where Benders-QAMOO secures the highest score of 824.3916.

Conversely, the exact classical solvers establish the absolute mathematical baseline for convergence at the expense of frontier diversity. The DPA anchors the true reference front, achieving a near-zero IGD of 0.0001, a minimal Spacing of 0.0665, and a near-perfect additive epsilon indicator (ϵ^+) of 0.0004. This mathematical precision is validated by the asymmetric Coverage indicators, where the exact classical sets achieve a absolute coverage ($C(B, A) = 1.0000$).

Second Notebook: The warm-start QAOA pipeline ($p = 2$ layers) successfully navigated the non-linear shock landscapes. Under the 600 MW collective AI data center surge, the quantum optimizer executed a 100% asset

reallocation shift from buses [8, 15] to [4, 10, 11, 12, 18], maintaining the absolute voltage floor safely above the critical 0.95 p.u. engineering limit. Across the full $N - 1$ contingency tree, the resilience-aware QAOA successfully minimized global continuous voltage deviations to a tight mean $|\Delta V|$ of 0.02705 p.u. across all valid outage conditions.

6 Quantum Platform Justification and Resource Needs

Requested: 156-qubit Heron-class QPUs (e.g., Kingston, Fez) (Inferior alternative: 120-qubit Eagle-class).

Our rationale for selecting Heron over Eagle:

- **Scale:** Heron supports up to 156 qubits, enabling direct variable encoding of grid cases up to 156 bus widths.
- **Fidelity:** Heron’s high 2-qubit gate fidelity ($> 99\%$) and low cross-talk are required for Samplomatic to resolve the signal (fails on noisier Eagle systems).
- **Integration:** Native support for Qiskit Executor primitives.

7 Gap Analysis and Implementation Roadmap

The following tasks are part of our Phase 3 roadmap:

- **True Quantum Hybrid Benders Loop:** Transition from the current offline classical preprocessor to an online iterative loop where the QPU directly executes the Master Problem QUBO solver in each iteration, executing the full Samplomatic error-mitigated pipeline (PEC + SLC + PNA) on IBM Heron.
- **Optimality Cuts:** Implement dual-based optimality cuts to bound and optimize objective function values, moving beyond feasibility-only cuts.

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